

Problem Set

Polynomials

- Let $P(x)$ be a monic polynomial of degree 5 such that $P(1) = 4$, $P(2) = 8$, $P(3) = 12$, $P(4) = 16$, and $P(5) = 20$. Find $P(7)$.
- Let $P(x)$ be a polynomial with integer coefficients such that $P(1) = 1$, $P(2) = 2$. Prove that 60 divides $P(10) - P(5) - 5$.
- Suppose a_1, a_2, a_3 are the roots of the polynomial equation $2x^3 + 3x^2 + 7x + 4 = 0$. Find the values of:
 - $\frac{1}{a_1} + \frac{1}{a_2} + \frac{1}{a_3}$
 - $a_1^3 + a_2^3 + a_3^3$
 - $(1 + a_1)(1 + a_2)(1 + a_3)$
 - $\frac{1}{1 + a_1} + \frac{1}{1 + a_2} + \frac{1}{1 + a_3}$
 - $a_1^4 + a_2^4 + a_3^4$
- Suppose all the roots of the equation

$$x^4 - 4x^3 + ax^2 + bx + 1 = 0$$

are positive real numbers. Show that all the roots are equal.

- Suppose two of the roots of the equation $x^4 - 8x^3 + ax^2 + bx + 12 = 0$ are 1 and 2. Find the other two roots.
- Suppose $\alpha_1, \alpha_2, \alpha_3$ are the roots of the polynomial equation $x^3 + ax^2 + bx + c = 0$. Find the equation whose roots are $\alpha_1^2 + 1$, $\alpha_2^2 + 1$, $\alpha_3^2 + 1$.
- Let α, β be the roots of the quadratic equation $ax^2 + bx + c = 0$. Find the quadratic equation whose roots are $\alpha^2\beta$, $\alpha\beta^2$.
- If α, β are the roots of $2x^2 - 5x - 4 = 0$, find the quadratic equation whose roots are $\alpha + \frac{1}{\alpha}$, $\beta + \frac{1}{\beta}$.
- If a, b are two of the roots of $x^4 + x^3 - 1 = 0$, prove that ab is a root of $x^6 + x^4 + x^3 - x^2 - 1 = 0$.
- Suppose u, v, w are the roots of the polynomial

$$P(x) = x^3 + 20x^2 + 13x + 42,$$

and $\frac{1}{u}, \frac{1}{v}, \frac{1}{w}$ are the roots of the polynomial

$$Q(x) = x^3 + rx^2 + sx + t.$$

Find $Q(1)$.

- α, β, γ are the roots of the equation $x^3 + px^2 + qx + r = 0$. Find the equation whose roots are

$$\alpha - \frac{1}{\beta\gamma}, \quad \beta - \frac{1}{\alpha\gamma}, \quad \gamma - \frac{1}{\alpha\beta}.$$

12. Let $P(x) = x^3 - 6x^2 + 11x - 6$ be a cubic polynomial with r_1, r_2, r_3 as its roots. Find the monic cubic polynomial whose roots are $\frac{3}{r_1+1}, \frac{3}{r_2+1}, \frac{3}{r_3+1}$.
13. Let $P(x)$ be a polynomial of degree 98 such that $P(k) = \frac{1}{k}$, for $k = 1, 2, \dots, 99$. Find $P(100)$.
14. Let a and b be two real numbers. If the roots of the quadratic equation

$$x^2 - ax - b = 0$$

are real and have absolute value less than 1, then show that each of the following holds:

- $|b| < 1$
 - $a + b < 1$
 - $b - a < 1$
15. Let $ax^2 + bx + c$ be a quadratic polynomial with real coefficients for which $|ax^2 + bx + c| \leq 1$ holds for $0 \leq x \leq 1$. Prove that $|a| + |b| + |c| \leq 17$. Give an example where the equality holds.
16. Let b, c be real numbers and suppose the roots of the equation $x^2 + bx + c = 0$ are both real and greater than 1. Prove that $b + c + 1 > 0$.
17. Find the remainder when x^{100} is divided by $x^2 - 3x + 2$.
18. If a, b, c are real numbers so that

$$x^3 + ax^2 + bx + c = (x^2 + 1)g(x)$$

for some polynomial g . Prove that $b = 1$ and $a = c$.

19. Prove that there cannot exist a polynomial $f(x)$ for which $xf(x-1) = (x+1)f(x)$ holds for all real x .
20. Suppose $f(x)$ is a polynomial with integer coefficients and there exists a positive integer k such that none of the integers $f(1), f(2), \dots, f(k)$ is divisible by k . Prove that $f(x)$ has no integer zeros.
21. Let $f(x) = ax^2 + bx + c$, where a, b and c are real numbers. Suppose $f(x) \neq x$ for any real x . Prove that there does not exist any real solution to the equation $f(f(x)) = x$.
22. Suppose $f(x) = c_0 + c_1x + c_2x^2 + \dots + c_nx^n = (2+x)^n$. If

$$\sum_{i=0}^n c_i = 81,$$

find the largest coefficient c_j of f .

23. Let $1, \alpha_1, \dots, \alpha_6$ be the distinct roots of $x^7 - 1$. Find the value of

$$(1 - \alpha_1)(1 - \alpha_2) \cdots (1 - \alpha_6).$$

24. If $a, b, c > 0$, is it possible that each of the polynomials $P(x) = ax^2 + bx + c$, $Q(x) = cx^2 + ax + b$, $R(x) = bx^2 + cx + a$ has two real roots?
25. If a, b, c are the roots of the equation $x^3 + px^2 + qx + r = 0$, then find the equation whose roots are ab, bc, ca .

26. Let f be a monic polynomial with integer coefficients and there are four distinct integers a, b, c, d for which

$$f(a) = f(b) = f(c) = f(d) = 12.$$

Prove that there is no integer k for which $f(k) = 25$.

27. Suppose the equation $ax^2 + bx + c = 0$ has a rational solution. If a, b, c are integers, not all zero, then prove that at least one of a, b, c is even.
28. Find the number of polynomials of the form $x^3 + ax^2 + bx + c$, which are divisible by $(x^2 + 1)$, and where a, b and c belong to $\{1, 2, \dots, 10\}$.
29. Prove that, if the roots of $x^2 + px + q$ are real, then the roots of $x^2 + px + q + (x + a)(2x + p) = 0$ will be real for every real number a .
30. Let $f(x) = x^4 + x^3 + x^2 + x + 1$. Find the remainder when $f(x^5)$ is divided by $f(x)$.
31. Let $p(x)$ be a polynomial with integer coefficients. Assume that $p(a) = p(b) = p(c) = -1$, where a, b, c are three distinct integers. Prove that $p(x)$ has no integral zeros.
32. Let n be an even positive integer, and let $p(x)$ be an n -degree polynomial such that $p(k) = p(-k)$ for $k = 1, 2, \dots, n$. First prove that $p(x) = p(-x)$ for all x real, and hence conclude that there is a polynomial $q(x)$ such that $p(x) = q(x^2)$.
33. Prove that there is no polynomial $P(x) = a_n x^n + a_{n-1} x^{n-1} + \dots + a_0$ with integer coefficients and of degree at least 1 with the property that $P(0), P(1), P(2), \dots$ are all prime numbers.
34. Let $f(x) = a_n x^n + a_{n-1} x^{n-1} + \dots + a_1 x + a_0$ be a polynomial of degree $n \geq 1$ with integer coefficients. Show that there are infinitely many positive integers m for which $f(m)$ is not prime.
35. Let the zeros m, n, k of $t^3 + at + b$ be rational. Prove that the roots of $mt^2 + nt + k$ are rational.
36. A polynomial $P(x)$ of the n th degree satisfies $P(k) = 2^k$ for $k = 0, 1, \dots, n$. Determine $P(n + 1)$.
37. Find the polynomial whose roots are the cube of the roots of the polynomial $t^3 + at^2 + bt + c$ (here a, b, c are constants).
38. A certain polynomial $p(x)$ when divided by $(x - a), (x - b), (x - c)$ leaves remainders a, b, c , respectively. What is the remainder when $p(x)$ is divided by $(x - a)(x - b)(x - c)$? [a, b, c are distinct.]
39. Let a, b, c be distinct integers. Show that the polynomial $(x - a)(x - b)(x - c) - 1$ cannot be factored into the product of two non-constant polynomials with integer coefficients.
40. Let $P(x), Q(x)$ and $R(x)$ be polynomials such that $P(x^5) + xQ(x^5) + x^2R(x^5)$ is divisible by $x^4 + x^3 + x^2 + x + 1$. Prove that $P(x)$ is divisible by $(x - 1)$.
41. First show that a real polynomial of odd degree must have at least one real root. Hence conclude that if a polynomial with real coefficients does not have any real root, it must have even degree. Using the result prove the following:
Let $f(x)$ and $g(x)$ be nonzero polynomials with real coefficients such that $f(x^2 + x + 1) = f(x)g(x)$. Show that $f(x)$ has even degree.
42. Let

$$f(x) = a_0 + a_1 x + \dots + a_n x^n$$

be a polynomial whose coefficients satisfy the conditions $0 \leq a_i \leq a_0$ ($i = 0, 1, \dots, n$). Let

$$(f(x))^2 = b_0 + b_1x + \dots + b_{n+1}x^{n+1} + \dots + b_{2n}x^{2n}.$$

Prove that

$$b_{n+1} \leq \frac{1}{2}(f(1))^2.$$

43. Suppose u and v are two real numbers such that u, v and uv are the three roots of a cubic polynomial with rational coefficients. Show that at least one root is rational.

44. If a and b are real and unequal, prove that the equation

$$(a-b)x^n + (a^2-b^2)x^{n-1} + \dots + (a^n-b^n)x + a^{n+1} - b^{n+1} = 0$$

has at most one real root.

45. Let a, b, c be three distinct integers and $P(x)$ is a polynomial with integer coefficients. Show that it is impossible to have $P(a) = b$, $P(b) = c$ and $P(c) = a$, simultaneously.

46. Suppose the polynomial $x^n + a_1x^{n-1} + a_2x^{n-2} + \dots + a_n$ can be factored into

$$(x + r_1)(x + r_2) \cdots (x + r_n),$$

where $r_1, r_2, \dots, r_n$ are real numbers. Prove that $(n-1)a_1^2 \geq 2na_2$.

47. Let $p_1, p_2, \dots, p_n$ be distinct integers and let $f(x)$ be the polynomial of degree n given by

$$f(x) = (x - p_1)(x - p_2) \cdots (x - p_n).$$

Prove that the polynomial

$$g(x) = (f(x))^2 + 1$$

cannot be expressed as the product of two non-constant polynomials with integer coefficients.

48. Let $ax^2 + bx + c$ be a quadratic polynomial with real coefficients for which $|ax^2 + bx + c| \leq 1$ for $0 \leq x \leq 1$. Prove that $|a| + |b| + |c| \leq 17$. Give an example for which the equality holds.

49. The equation $(x - a_1)(x - a_2) \cdots (x - a_n) = 1$ where a_i 's are real numbers, has n real roots $r_1, r_2, \dots, r_n$. Find the minimum number of real roots of the equation

$$(x - r_1)(x - r_2) \cdots (x - r_n) = -1.$$

50. Let $f(x)$ be a polynomial which satisfies

$$f(x^2) + f(x)f(x+1) = 0.$$

Prove the results:

- (a) If r is a root of f , then so are r^2 and $(r-1)^2$.
 - (b) Hence show that only possible roots of f are 0 or 1.
 - (c) Finally, determine a general form of $f(x)$.
51. Let the polynomial $f(t) = t^n + a_{n-1}t^{n-1} + \dots + a_1t + 1$ have non-negative coefficients and n real zeros. Prove that $f(2) \geq 3^n$.
52. $p(x)$ and $q(x)$ are polynomials which satisfy the identity $p(q(x)) = q(p(x))$ for all real x . If the equation $p(x) = q(x)$ has no real solution, show that the equation $p(p(x)) = q(q(x))$ has no real solution.

53. Prove that the polynomial $a_n t^n + a_{n-1} t^{n-1} + \cdots + a_3 t^3 + t^2 + t + 1$ with all real coefficients cannot have all its zeros real.
54. For real numbers a, b, c , suppose that $29c + 10b + 2a = 0$. Then prove that the equation $ax^3 + bx + c = 0$ has at least one real root between 0 and 1.
55. Let $f(t)$ be any polynomial with real coefficients and with degree exceeding 1. Observe from the graph of f that there is a real value of k for which not all zeros of $f(t) + k$ are real (You don't need to prove this analytically. Just visualize the graph of f and recall that the operation $+k$ is nothing but shifting the graph of f by k amount upward or downward, depending on the sign of k). Use this observation in the following problem:
Let $p(t)$ be a polynomial for which $p(t)$ is real when t is real, and non-real when t is non-real. Prove that $p(t)$ must be linear. Approach via the steps given below:
- First show that $p(t)$ is real if and only if t is real.
 - Then prove that, the above statement also holds true for $p(t) + k$, for any real k .
 - Use (b) to prove that each root of $p(t) + k$ is real, for any real k .
 - Finally, conclude that, degree of $p(t)$ cannot exceed 1. And since $p(t)$ cannot be constant polynomial (why?), it must be linear.
56. Let $P(x) = a_0 x^n + a_1 x^{n-1} + \cdots + a_{n-1} x + a_n$ with $n \geq 3$. Suppose that $P(x)$ has all real roots and $a_0 = 1$, $a_1 = -n$, $a_2 = \frac{n^2 - n}{2}$. Determine coefficients a_i for $i = 3, \dots, n$.
57. Find all polynomials $p(x)$ with $xp(x-1) = (x-26)p(x)$ for all x .
58. The polynomial $ax^2 + bx + c$ with $a > 0$ has real zeros x_1, x_2 . Show that

$$|x_i| \leq 1 \ (i = 1, 2) \iff a + b + c \geq 0, \ a - b + c \geq 0, \ a - c \geq 0.$$

59. Find all polynomials $p(x)$ which satisfies

$$p(x+1) = p(x) + 2x + 1$$

for all x .

60. Suppose that a polynomial $P(x) = x^{10} - 10x^9 + 39x^8 + a_7 x^7 + \cdots + a_1 x + a_0$ has 10 real roots. Prove that all the roots of $P(x)$ are between -2.5 and 4.5 .
61. Let $P(x, y)$ be a symmetric polynomial in x and y , i.e. $P(x, y) = P(y, x)$. Suppose $(x - y)$ is a factor of $P(x, y)$ [i.e. $P(x, y) = (x - y)Q(x, y)$, for some polynomial $Q(x, y)$]. Prove that $(x - y)^2$ is a factor of $P(x, y)$.
62. Let $ax^2 + bx + c$ be a quadratic polynomial with real coefficients which satisfies $|ax^2 + bx + c| \leq 1$ for $|x| \leq 1$. Show that $|2ax + b| \leq 4$ whenever $|x| \leq 1$.
63. If $x^3 + px^2 + qx + r$ has three real zeros, prove that $p^2 \geq 3q$.
64. Let a, b, c be real numbers. Consider the equation

$$(x - a)(x - b) + (x - b)(x - c) + (x - c)(x - a) = 0.$$

Prove that the roots of this equation are always real. Further show that the roots are equal if and only if $a = b = c$.

65. The polynomial $ax^3 + bx^2 + cx + d$ has integral coefficients with ad odd and bc even. Prove that at least one zero of the polynomial is irrational.